

Interim Update: Architecture Order and eRank Dynamics in Modular-Addition Transformers

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INTERIM REPORT — 80 of 102 runs complete; 8 still running; 14 pending.

1 Executive Summary

- This is an *interim* snapshot of a single-seed sweep over all balanced permutations of attention (A) and MLP (M) blocks at depths 2, 3, 4 on modular addition mod 113. Out of 102 scheduled runs (96 balanced + 6 unbalanced controls), 80 have a terminal status (76 GROKKED + 4 MEMORIZED_ONLY), 8 are still training, and 14 are still in the queue.
- Among the 80 balanced runs that have completed, **Pearson**(a.before.m.score, test_acc) = **0.479** but **Spearman** = **-0.033** — the Pearson is driven by a small number of low-score schedules that fail; among the schedules that grok, ordering does not monotonically predict accuracy.
- The strongest pattern is in **first-block identity**, not in the overall ordering score. **Every completed schedule starting with A has grokked** (48/48 = 100%, mean test accuracy 0.998). **Schedules starting with M grok 28/32 = 87.5%** (mean test accuracy 0.878). The 4 MEMORIZED_ONLY schedules are all M -first.
- **eRank compression is a strong predictor of grokking among completed runs:** Pearson(final_rep.eRank, test_acc) = -0.887. Same-sum cluster ratio (lower = better task-aligned geometry) shows -0.799.
- Componentwise Δ eRank reveals the mechanism for M -first failure cleanly: in MMAA and similar schedules, the leading M blocks produce Δ eRank ≈ 0 , indicating that an MLP cannot expand a rank-1 representation when it has not yet seen position-dependent information from attention.
- Depth-4 schedules appear more tolerant of unfavorable ordering than depth-2 schedules: of the 23 depth-4 schedules completed so far, 23/23 have grokked, including schedules with low routing-before-processing score.
- These are single-seed observations and should be treated as suggestive. Re-running the strongest and weakest schedules across seeds 1 and 2 is the most important next step.

2 Experimental Design

Task. Modular addition $c = (a + b) \bmod 113$. Inputs are length-3 sequences $[a, b, =]$; the model predicts c at position 2 over a vocabulary of size 113.

Model. Custom `BlockScheduleTransformer` ($d_{\text{model}} = 128$, 4 heads, $d_{\text{head}} = 32$, $d_{\text{mlp}} = 512$, GELU MLPs, learned positional embeddings, no LayerNorm, causal mask). Every block is either an attention-only update $R^{(k+1)} = R^{(k)} + A_k(R^{(k)})$ or an MLP-only update $R^{(k+1)} = R^{(k)} + M_k(R^{(k)})$.

Schedules. For each depth $d \in \{2, 3, 4\}$, every unique balanced permutation of d attention blocks and d MLP blocks:

$$\binom{4}{2} = 6, \quad \binom{6}{3} = 20, \quad \binom{8}{4} = 70.$$

Plus six unbalanced controls (all- A and all- M at each depth). Total: $96 + 6 = 102$ schedules. Seed = 0 only.

Training. AdamW lr = $3 \cdot 10^{-4}$, weight decay 1.0, batch 256. Validation-selected best checkpoint; test accuracy evaluated only at the best-val checkpoint. Maximum 50,000 epochs with early stopping if validation accuracy ≥ 0.95 for 200 consecutive evals.

Routing-before-processing score. For a schedule of length $2d$ over blocks $C_1, C_2, \dots, C_{2d}$, define

$$S_{\text{A before M}} = \frac{1}{d^2} \sum_{i: C_i=A} \sum_{j: C_j=M} \mathbf{1}[i < j].$$

$S = 1$ means all A 's precede all M 's (AAMM, AAAMMM, AAAAMMMM). $S = 0$ means the reverse. $S \approx 0.5$ is interleaved.

eRank. For an activation matrix $X \in \mathbb{R}^{n \times d}$ (column-centered) with singular values σ_i , the effective rank is

$$\text{eRank}(X) = \exp\left(-\sum_i \frac{\sigma_i}{\sum_j \sigma_j} \log \frac{\sigma_i}{\sum_j \sigma_j}\right),$$

following Roy and Vetterli (2007). Componentwise deltas are $\Delta^{(k)} = \text{eRank}(R^{(k+1)}) - \text{eRank}(R^{(k)})$.

3 Current Run Status

Table 1: Run status by depth (interim).

Depth	Total	Complete	Grokked	Memorized	Failed-to-fit	Running	Pending
2	8	6	4	2	0	0	2
3	22	20	18	2	0	0	2
4	72	54	54	0	0	8	10
Total	102	80	76	4	0	8	14

The remaining 22 pending/running schedules are concentrated in the depth-4 **MMAA/MMAAM**-prefix region (low routing-before-processing score). Depth-2 controls (**AAAA, MMMM**) and depth-3 controls (**AAAAAA, MMMMMM**) are still pending.

4 Accuracy and Grokking Results

4.1 Top 10 schedules by test accuracy (completed runs only)

Table 2: Top 10 completed schedules by test_acc@best_val. All grokked.

Rank	Depth	Schedule	S	alt	best_val	test@best_val	τ	status
1	4	AAMMAAMM	0.75	3	1.0000	1.0000	–	GROKKED
2	4	AMAMMAAM	0.56	5	1.0000	1.0000	–	GROKKED
3	3	MAAMAM	0.56	4	1.0000	0.9999	4075	GROKKED
4	4	AAAMAMMM	0.94	3	1.0000	0.9998	–	GROKKED
5	4	AAAMMAMM	0.88	3	1.0000	0.9998	–	GROKKED
6	3	MAMAMA	0.33	5	1.0000	0.9998	7350	GROKKED
7	2	AMAM	0.75	3	1.0000	0.9994	7025	GROKKED
8	3	MAMAAM	0.44	4	1.0000	0.9994	3125	GROKKED
9	4	AMAAMMMA	0.62	4	1.0000	0.9991	–	GROKKED
10	4	AMAMAMAM	0.62	7	1.0000	0.9991	–	GROKKED

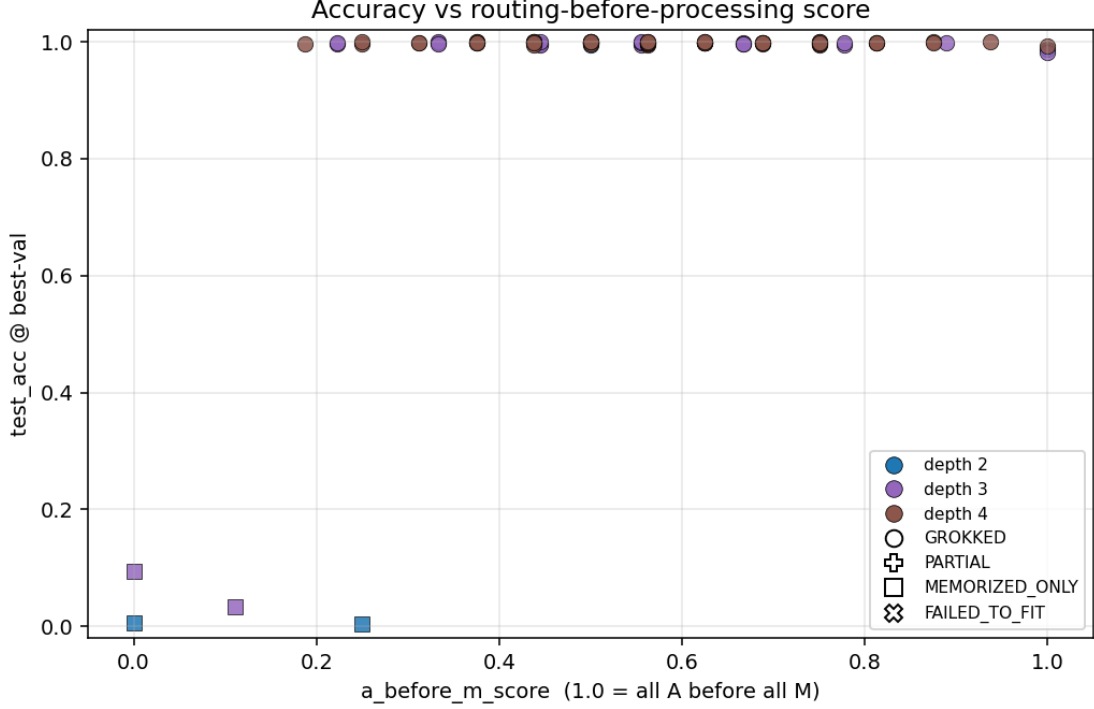


Figure 1: Completed balanced runs only. Test accuracy at best-validation checkpoint vs. S_A before M . Color by depth, marker by status. The $S = 0$ corner is the only region producing **MEMORIZED_ONLY** outcomes.

Table 3: Lowest-performing completed schedules.

Depth	Schedule	S	alt	best_val	test@best_val	status
2	MAMA	0.25	3	0.009	0.004	MEMORIZED_ONLY
2	MMAA	0.00	1	0.017	0.006	MEMORIZED_ONLY
3	MMAMAA	0.11	3	0.014	0.012	MEMORIZED_ONLY
3	MMMAAA	0.00	1	0.013	0.013	MEMORIZED_ONLY
3	AAAMMM	1.00	1	1.000	0.981	GROKKED
2	AAMM	1.00	1	0.995	0.987	GROKKED

4.2 Bottom 10 / failed schedules (completed runs only)

The 4 **MEMORIZED_ONLY** schedules so far are all depth 2 or 3 and all start with M . No depth-4 schedule has failed in the completed subset.

4.3 Correlations

Interpretation. The geometry correlations (eRank, same-sum) are much stronger than the ordering correlation, suggesting that block ordering matters mostly through whether it enables the model to compress into a task-aligned geometry. The Spearman of -0.03 for ordering vs. accuracy shows that the Pearson is driven by a few extreme failures: among the 76 schedules that grokked, ordering does not predict accuracy ranking.

4.4 Routing-before-processing buckets

All four observed failures sit in the *low* bucket. Both the middle and high buckets are 100% grokked in the completed subset.

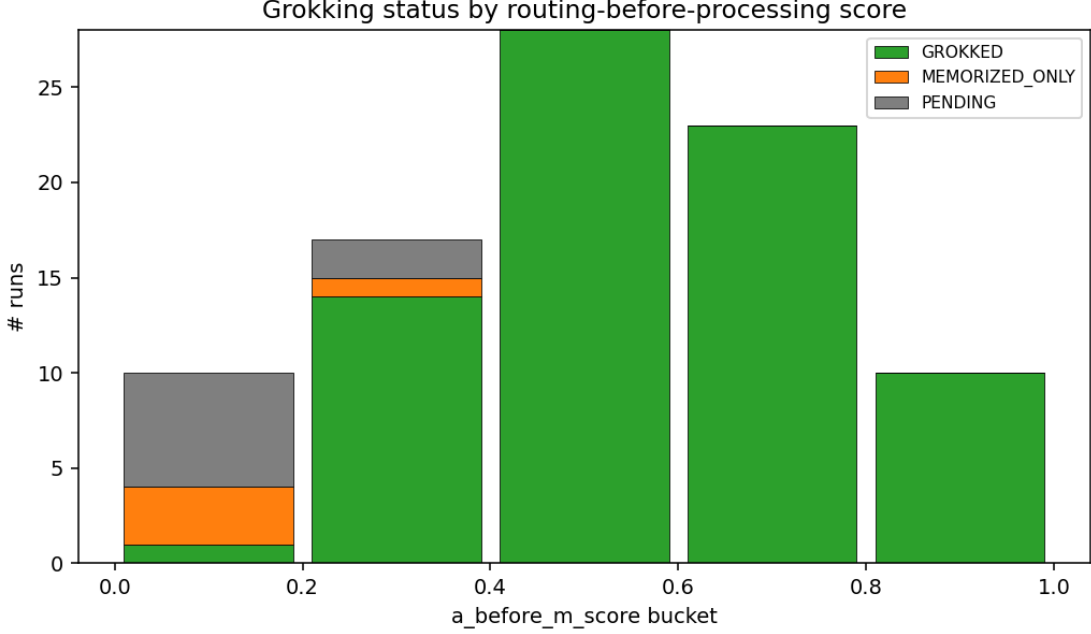


Figure 2: Status counts in five buckets of $S_{A \text{ before } M}$. MEMORIZED_ONLY runs concentrate at low S ; high- S buckets are entirely GROKKED in the completed subset.

Table 4: Correlations across $n = 80$ completed balanced schedules.

Pair	Pearson	Spearman
$S_{A \text{ before } M}$ vs. test_acc	+0.479	−0.033
num_alternations vs. test_acc	+0.281	+0.336
final_rep_eRank vs. test_acc	− 0.887	—
top-5 singular mass vs. test_acc	+0.500	—
same-sum ratio vs. test_acc	− 0.799	—

4.5 First and last block

First-block identity is the cleanest single binary predictor of grokking. Every A -first schedule has grokked; the four observed failures are all M -first. Mirror image at the end: every M -last schedule has grokked; the failures are A -last.

5 Componentwise eRank

For a schedule of length $K = 2d$, the residual stream evolves

$$R^{(0)} \rightarrow R^{(1)} \rightarrow \dots \rightarrow R^{(K)},$$

and $\Delta^{(k)} = \text{eRank}(R^{(k+1)}) - \text{eRank}(R^{(k)})$ attributes the per-block representational change.

5.1 Representative per-block Δ -eRank values

Mechanistic interpretation (tentative). Before any A block has acted, every position in the residual stream depends only on the (positional + token) embedding of its own input, and the per-position activation distribution across the batch is effectively rank-1 in the dimension that matters for prediction. An MLP applied positionwise to a stream with this structure cannot expand its eRank, because it acts identically on each batch row (modulo the constant positional embedding). This is exactly what we observe: M -only blocks before any A have Δ -eRank within numerical noise of zero. Once an attention block runs, the stream becomes a function of all input positions

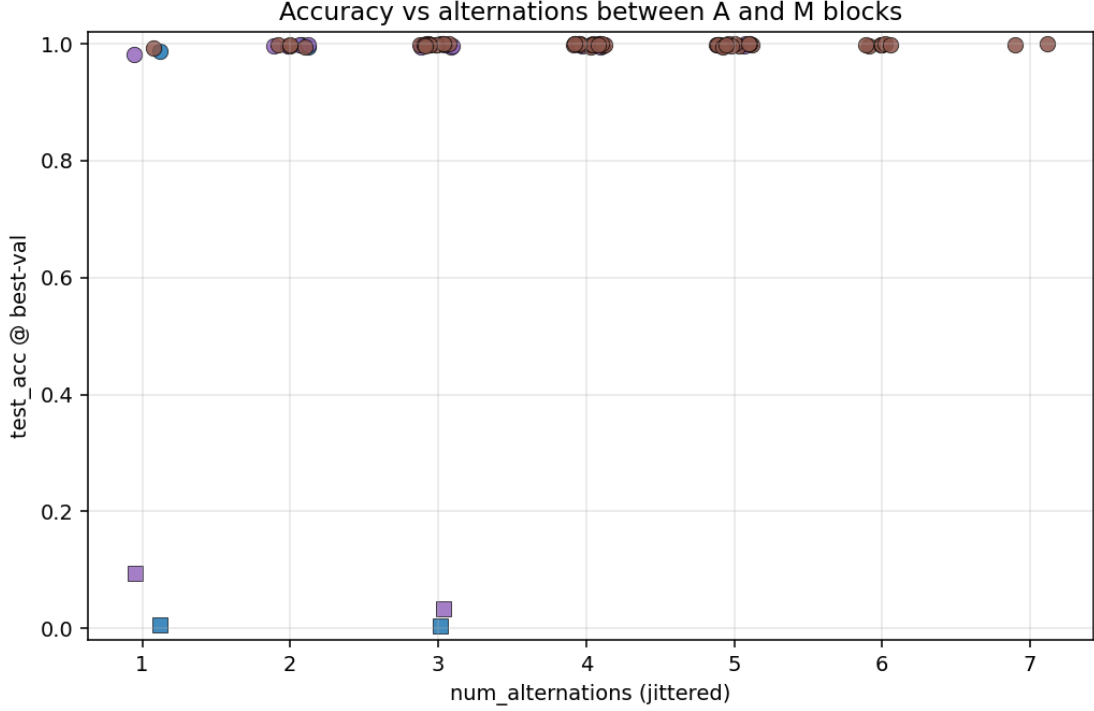


Figure 3: Test accuracy vs. number of $A \leftrightarrow M$ transitions in the schedule. Among completed runs, alternation count is weakly positively correlated with accuracy.

Table 5: Completed balanced schedules grouped by S_A before M .

Bucket	S range	n	mean test_acc	% grokked
Low	≤ 0.33	11	0.648 (sd 0.464)	63.6%
Middle	$(0.33, 0.67)$	46	0.998 (sd 0.002)	100%
High	≥ 0.67	23	0.997 (sd 0.004)	100%

and subsequent MLP blocks become useful. Schedules that defer attention pay the cost twice: they waste their leading M blocks, and they crowd the remaining A blocks into a smaller depth budget for routing.

6 Training-Time Dynamics and Compression

Caveat on geometric metrics. Δ eRank, top- k mass, and same-sum clustering all show strong correlation with test accuracy among completed runs, but eRank changes are geometric and need not reflect task-relevant computation. The memorized schedules retain non-trivial eRank in their final A blocks even though they fail to generalize, reinforcing that geometric compression is necessary-looking but not sufficient by itself.

7 Potential Findings (tentative, single-seed)

1. **First-block identity is the cleanest predictor of grokking observed so far.** Every A -first schedule has grokked; every observed failure is M -first. The M -first failure rate in the completed subset is $4/32 = 12.5\%$.
2. **Routing-before-processing schedules do not need to be extreme to succeed.** The middle bucket ($0.33 < S < 0.67$, e.g. schedules like AMAM and MAMAMA) groks 100% of the time and produces some of the highest test accuracies. Strict AAAMMM-style ordering ($S = 1$) is not required.

Table 6: Outcome by first and last block identity (completed balanced schedules).

First block	n	mean acc	% grokked	Last block	n	mean acc	% grokked
A	48	0.998	100%	A	37	0.894	89%
M	32	0.878	87.5%	M	43	0.998	100%

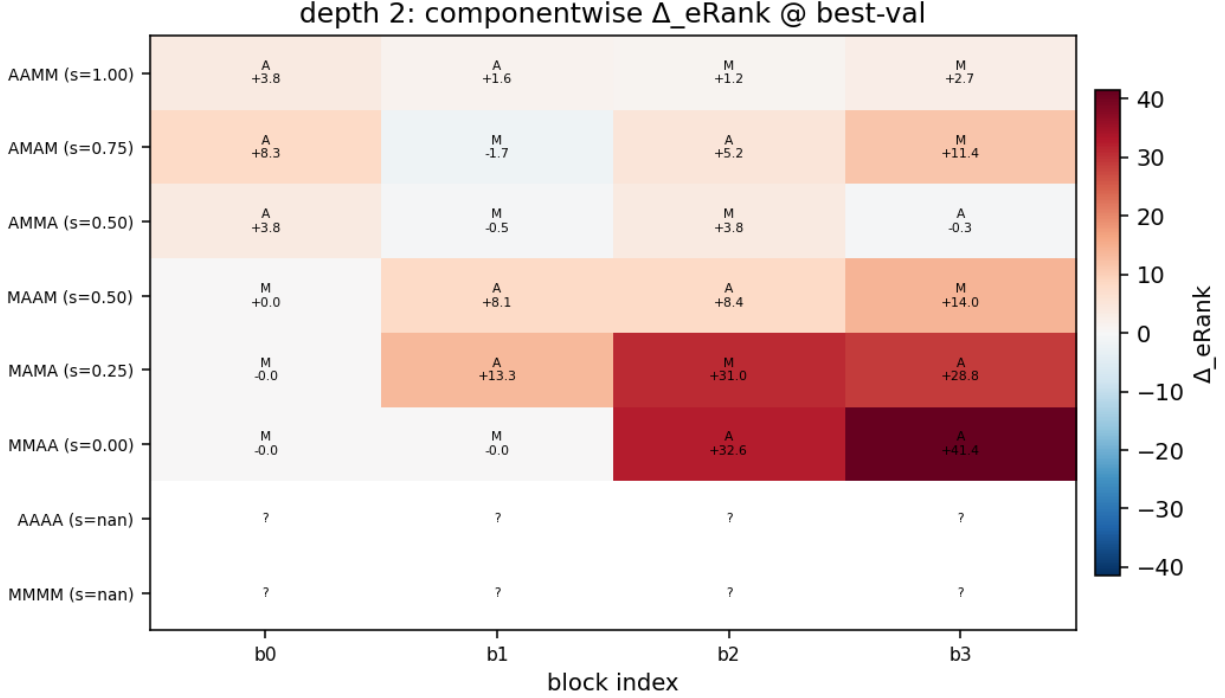


Figure 4: Depth-2 (4-block) componentwise Δ_{eRank} at best-val. Rows sorted by S_A before M (high at top). The failed schedules MMAA and MAMA show $\Delta \approx 0$ in their leading M blocks.

- M -blocks before any A produce no representational change.** The leading M blocks of MMAA, MAMA, MMMAAA, etc. show $\Delta_{\text{eRank}} \approx 0$ at best-val. This is a mechanistic signature: an MLP on a position-symmetric rank-1 stream cannot expand its rank.
- eRank compression is consistent with grokking but not specific to it.** Strongly grokked schedules tend to have low final-rep eRank and high top-5 singular mass. Failed schedules can still have non-trivial eRank growth in their final A blocks — compression is necessary-looking but not sufficient by itself.
- Depth dilutes the ordering effect.** Depth-4 schedules grok 100% (54/54) in the completed subset, including some low-score orderings. The depth-2 MMAA/MAMA failures are the cleanest demonstrations of the routing-before-processing constraint; depth 3 and 4 appear more forgiving. The remaining 18 depth-4 schedules (low-score region) will refine this estimate.

8 Caveats

- Interim.** 80/102 runs are complete; the remaining 22 are concentrated in the low- S depth-4 region. If those runs reveal late depth-4 failures, the depth-tolerance finding would weaken.
- Single seed (seed = 0).** No multi-seed reliability check.
- Architectures are not parameter-matched.** A-only blocks have no MLP; M-only blocks have no attention. At $d_{\text{model}} = 128, d_{\text{mlp}} = 512$: AAAA has 294k params, MMMM has 556k, AMAM-style schedules have 425k. Capacity differences confound architecture with ordering.

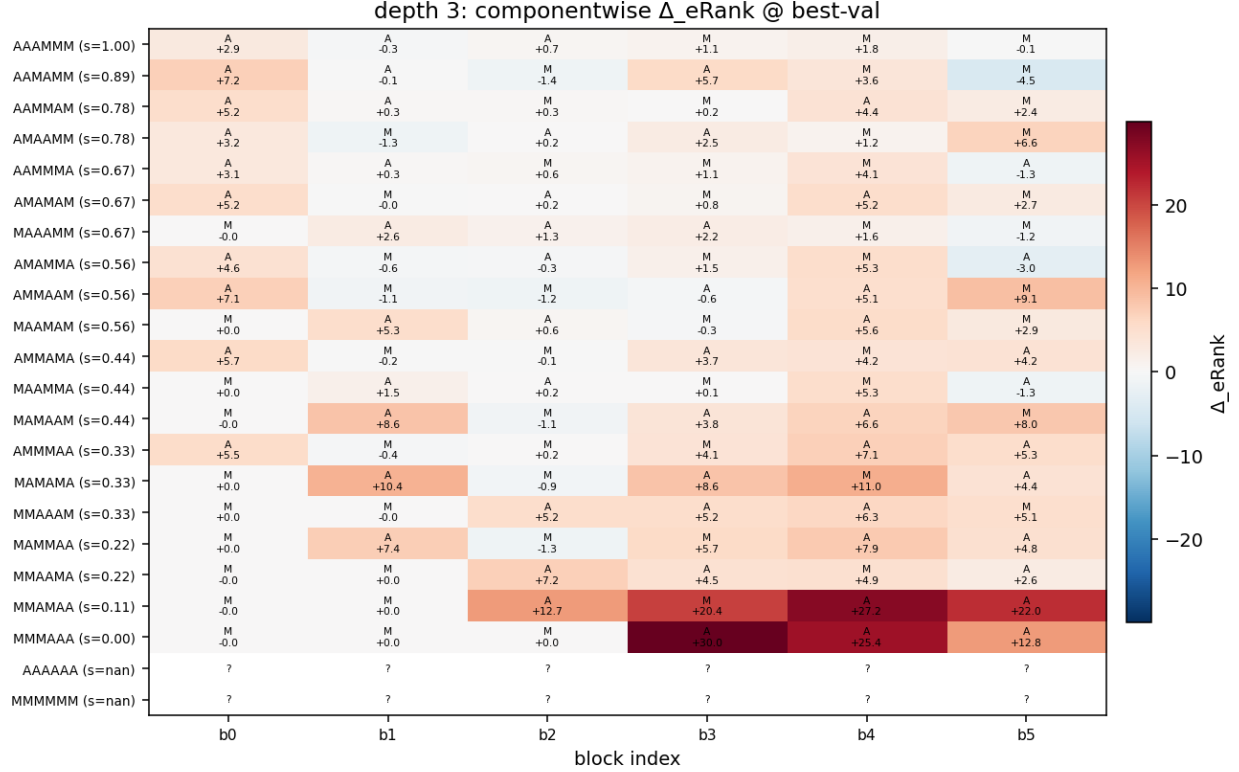


Figure 5: Depth-3 componentwise Δ_{eRank} at best-val. Same pattern: leading M blocks contribute essentially zero geometric change. Compensating A -block contributions in later positions are large.

- **Failed-model eRank is not successful-circuit geometry.** The geometric metrics for MMAA, MAMA, etc. describe failed models and should not be over-interpreted as evidence about the underlying task structure.
- **Geometric correlation $\neq$ causation.** eRank, top- k mass, and same-sum clustering all correlate with test accuracy at best-val, but the direction of dependence is not established from this data alone.
- **Custom transformer.** The BlockScheduleTransformer is not HookedTransformer; it mirrors the modular-addition baseline (no LayerNorm, GELU MLPs, learned positional embeddings, causal mask) but small numerical differences from prior HookedTransformer-based runs are possible.
- **Unbalanced controls (all-A and all-M at each depth) are not yet completed.** They are currently pending and will be most informative as baselines once they run.

9 Next Steps

1. Finish the remaining 22 runs (mostly depth-4 low- S schedules + all 6 controls).
2. Re-run the top-3 grokking schedules and the bottom-3 failing schedules with seeds 1 and 2 to confirm single-seed observations.
3. Parameter-match a small subset: replicate AAMM vs. MMAA with d_{mlp} tuned so that the two architectures have identical total parameter counts, to disentangle ordering from capacity.
4. Investigate the depth-4 mid- S schedules that grok despite low alternation count (e.g. AAAMAMMM, AAAMMAMM) — these are candidates for further mechanistic study of the late-attention pattern.
5. Add denser training-time checkpointing around the grokking phase transition for the strongest schedule at each depth, to localize when the final-rep eRank compresses.

Table 7: Per-block Δ eRank at the best-validation checkpoint, for representative schedules. Block-index 0 = first block.

Schedule	Status	Δ eRank per block (A/M)
AAMM	GROKED	$A_0=+3.83, A_1=+1.58, M_2=+1.19, M_3=+2.66$
AMAM	GROKED	$A_0=+8.27, M_1=-1.74, A_2=+5.16, M_3=+11.36$
MAMA	MEMORIZED_ONLY	$M_0=+0.00, A_1=+13.28, M_2=+30.99, A_3=+28.82$
MMAA	MEMORIZED_ONLY	$M_0=+0.00, M_1=+0.00, A_2=+32.65, A_3=+41.43$
AAAMMM	GROKED	$A_0=+2.92, A_1=-0.31, A_2=+0.70, M_3=+1.13, M_4=+1.77, M_5=-0.15$
MMMAAA	MEMORIZED_ONLY	$M_0=+0.00, M_1=+0.00, M_2=+0.00, A_3=+29.98, A_4=+25.41, A_5=+12.82$
AAAAAMMM	GROKED	$A_0=+9.60, A_1=-0.17, A_2=-0.50, A_3=+0.54, M_4=+0.16, M_5=+0.92, M_6=+5.66, M_7=+5.66$

6. Once the unbalanced controls (AAAA, MMMM, etc.) finish, compare them as parameter-mismatched but pure-component baselines.

Acknowledgements / sources

Source data and aggregation scripts: see `outputs/arch_order_sweep_6gpu/` and `scripts/aggregate_arch_order_sweep.py`.
Full implementation handoff: `/jumbo/lisp/f006j44/erank/arch_order_sweep_implementation.md`.

depth 4: componentwise $\Delta_e\text{Rank}$ @ best-val

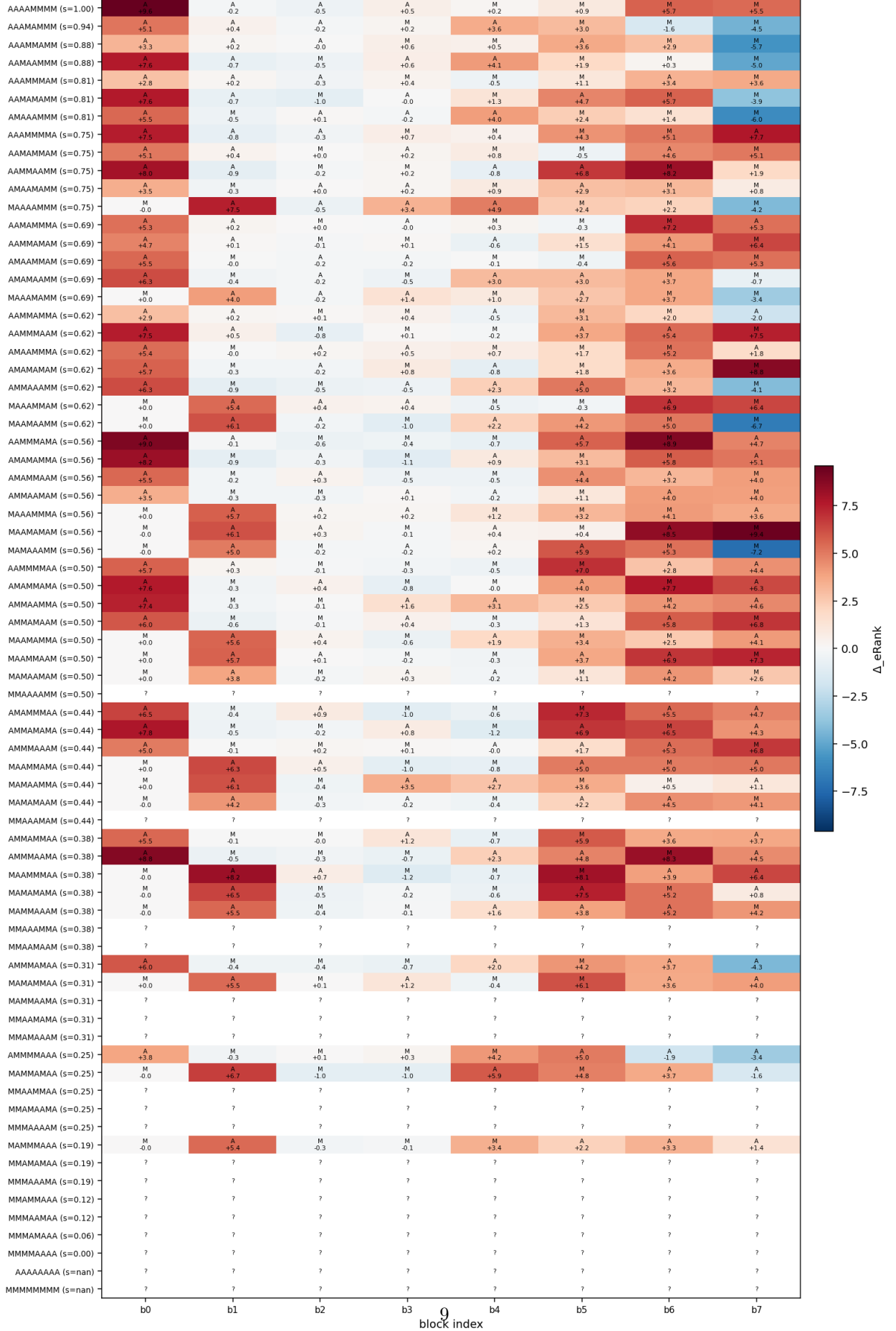


Figure 6: Depth-4 componentwise $\Delta_e\text{Rank}$ for the 54 completed schedules. Leading- M schedules in the bottom

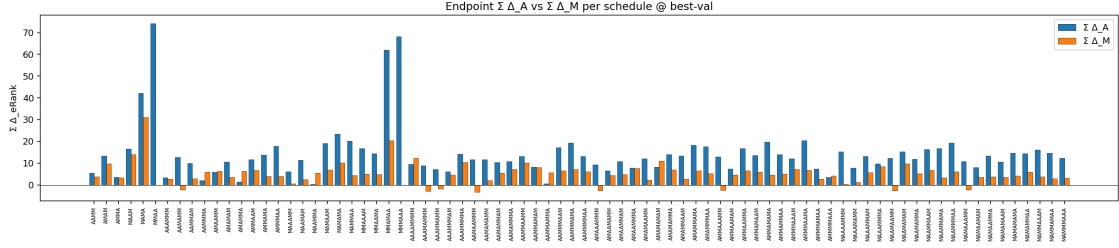


Figure 7: $\sum_A \Delta_eRank$ and $\sum_M \Delta_eRank$ per schedule. M -first schedules need very large $\sum_A$ near the output to compensate for the empty M blocks at the input side.

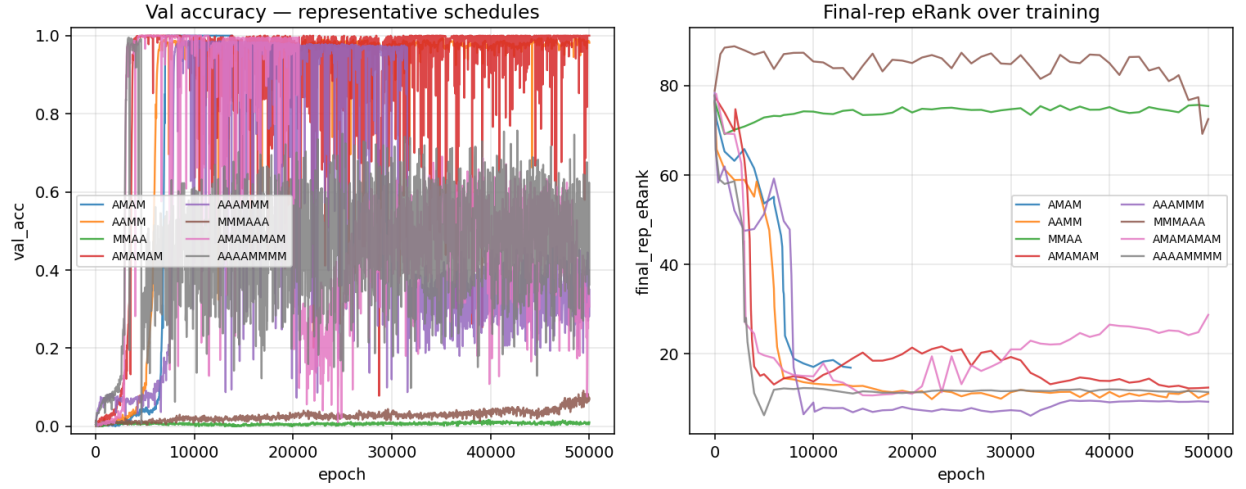
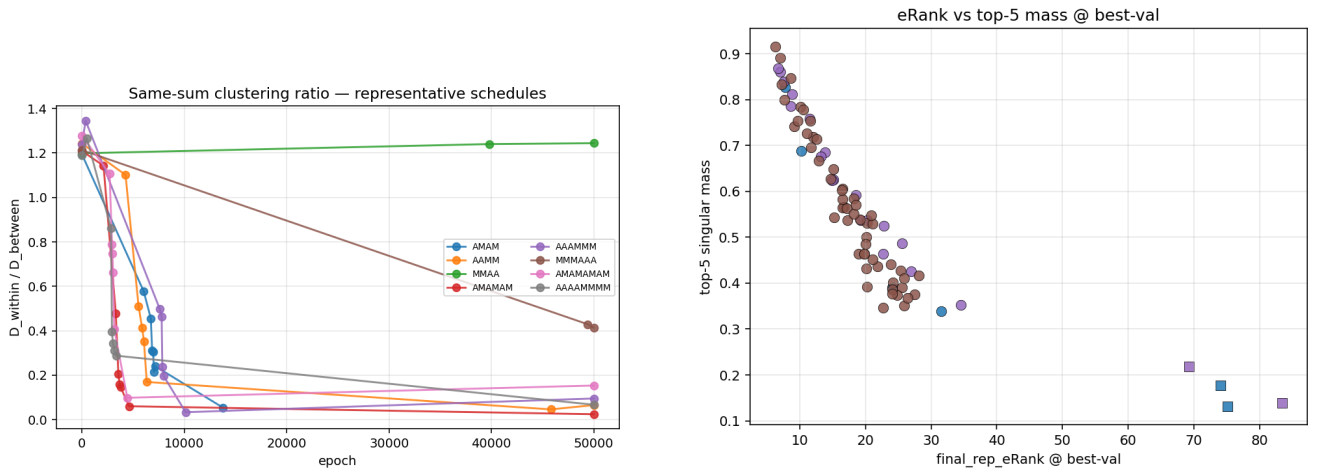


Figure 8: Validation accuracy and final-residual eRank over training, for representative schedules. Grokking is accompanied by a fall in final-rep eRank in successful schedules. Memorized schedules retain higher eRank without generalizing.



(a) Same-sum within/between cluster ratio over training for representative schedules. Lower = more task-aligned geometry.

(b) Final-rep eRank vs. top-5 singular mass at best-val. Compression and spectral concentration both correlate with grokking.

Figure 9: Spectral compression and task-aligned clustering accompany grokking.